

Intersection of lines and planes

Key Points

The intersection of a line with a plane may be a single coordinate or the line may be in-planar or it may not intersect at all

To solve any intersection all coordinates must be equal for given parameters

Basic Skills

Q1. Write down the coordinates x, y, z in terms of the parameters for the equation of the lines and planes

a) $r = \begin{pmatrix} 5 \\ -2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$ **b)** $r = \begin{pmatrix} 7 \\ 0 \\ -3 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$

c) $r = \begin{pmatrix} 5 \\ -2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} - \mu \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix}$

Q2. Find the cartesian equation of $\left(r - \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix}\right) \cdot \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} = 0$

Q3. Find the point of intersection between the lines

$r_1 = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} + \lambda \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$ and $r_2 = \begin{bmatrix} -4 \\ 0 \\ 8 \end{bmatrix} + \mu \begin{bmatrix} 6 \\ -10 \\ -3 \end{bmatrix}$, if it exists

Competency Skills

Q4. For the line $r = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 5 \end{pmatrix}$ and the plane $2x - y + z = 13$

a) Find $x(\lambda), y(\lambda), z(\lambda)$

b) By substitution show that $\lambda = 1$ and hence find the intersection between the line and plane

Q5. Find the point at which the line $r = \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix}$ intersects the plane $3x + 2y + z = 16$

Q6. Show that the line $r = \begin{pmatrix} 5 \\ 1 \\ -1 \end{pmatrix} + t \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}$ is contained in the plane $y + 6z = 10 - 3x$

Q7. Find where $r = \begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ intersects the plane with normal $n = (2, 5, -1)$ going through the origin

Advanced Skills

Q8. The line l_1 goes through the point A $(2, 3, -5)$ and goes in the direction parallel to the line

described by $r = \begin{bmatrix} -2 \\ 1 \\ 6 \end{bmatrix} + t \begin{bmatrix} -4 \\ 0.5 \\ 1 \end{bmatrix}$, find out when the line l_1 intersects the plane given by $3x + y - z = -5$

Q9. Determine the line of intersection between planes with normals $n = (5, 2, 0)$ and $n = (1, 2, -1)$ both going through the origin

Q10. Find k , if it exists, such that the line $r = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix} + t \begin{bmatrix} 2k \\ 1 \\ 5 \end{bmatrix}$ is contained in the plane $x + y - z = 1$

Extension

Q11. Explain using the theory of simultaneous equations why the points that a plane and line intersect are either 0, 1 or infinitely many